SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title: AI-Powered Crew Assembly for Event Staffing

Team Name: TwinTech

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Division: SE-C2(2nd year)

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1. Introduction

The AI-Powered Crew Assembly system helps event organizers find and assemble suitable staff such as photographers, decorators, sound engineers, anchors, and security personnel. It matches candidates based on skills, availability, location, budget, and reliability.

2. Purpose

This document describes the functional and non-functional requirements of the system. It guides developers and testers in designing and implementing an efficient event staffing solution.

3. Scope

- Organizers can create events and specify crew requirements.
- Freelancers can register and maintain their profiles.
- AI matches candidates according to event requirements.
- The system provides backup crew suggestions for cancellations.
- Organizers can manage assignments and review performance.

4. Definitions, Acronyms, and Abbreviations

- AI: Artificial Intelligence used for intelligent crew matching.
- Crew: Freelancers working at an event.
- Organizer: Person responsible for event planning.
- Backup Crew: Alternative staff available for replacement.

5. Overall Description

5.1 Users and Characteristics

User Type	Description
Organizer	Creates events and selects crew.
Freelancer	Manages profiles and accepts assignments.
Admin	Manages users and monitors the system.

5.2 Assumptions and Dependencies

- The system is accessible through web browsers and mobile devices.
- Internet connectivity is required.
- Freelancers must maintain updated profiles and availability.
- Secure login is implemented.

6. Functional Requirements

6.1 User Registration/Login

- Register and log in securely.
- Reset forgotten passwords.

6.2 Event Management

- Add, edit, and delete event details.
- Specify date, location, budget, and required roles.

6.3 Freelancer Management

- Add skills, experience, location, and availability.
- Update profiles and accept or reject assignments.

6.4 AI-Based Crew Matching

- Match freelancers based on skills, availability, location, budget, and reliability.
- Display suitable candidates for organizer approval.

6.5 Backup Crew Management

- Suggest alternative candidates for important roles.
- Help replace freelancers who cancel.

6.6 Notifications and Analytics

- Notify users about assignments and cancellations.

- Generate reports on staffing, cancellations, and performance.

7. Non-Functional Requirements

- Performance: Candidate searches should respond quickly.
- Security: User authentication and data protection.
- Usability: Simple and user-friendly interface.
- Scalability: Support multiple users and events.
- Reliability: Maintain accurate event and availability information.

8. System Architecture

- Frontend: React for the user interface.
- Backend: Node.js and Express for application logic.
- Database: MongoDB for storing user and event data.
- Matching Engine: AI-assisted or rule-based candidate matching.
- Authentication: Secure user login and access control.
- Notifications: Email or notification services.

9. Constraints

- Matching depends on accurate freelancer information.
- Internet connectivity is required.
- User data must remain secure.
- Recommendations must satisfy event requirements and budget limits.
- Organizers make the final selection.

10. Future Enhancements

- Mobile application development.
- Automated payment and invoice generation.
- Calendar integration for availability.
- AI-based cancellation risk prediction.
- Advanced staffing analytics.

Feasible Solution

Technical Solution and Plan

- Frontend: React
- Backend: Node.js and Express
- Database: MongoDB
- Matching: AI-assisted matching algorithm
- Authentication: Secure login system
- Hosting: Cloud platform

Implementation Steps

1. Requirements finalization.
2. UI/UX design.
3. Database design.
4. Module development.
5. Testing.
6. Deployment and monitoring.

Tools and Technologies

React, Node.js, Express, MongoDB, GitHub, REST APIs, and optional AI APIs.